

Chronic toxicity and carcinogenicity of dietary administered ammonium sulfate in F344 rats

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Received 10 February 2005; accepted 1 June 2005

Abstract

Chronic toxicity and carcinogenicity studies of ammonium sulfate, used as a food additive in fermentation, were performed in male and female Fisher 344 rats at dietary concentrations of 0%, 0.1%, 0.6% and 3.0% in a 52 week toxicity study and 0%, 1.5% and 3.0% in a 104 week carcinogenicity study. Treatment with ammonium sulfate caused significant increase in kidney and/or liver weights in males and females of the 3.0% diet group, but no effects were found on survival rate, body weights, and hematological, serum biochemical or histopathological parameters at any dose levels in the chronic toxicity study. Regarding carcinogenicity, ammonium sulfate did not exert any significant influence on the incidences of tumors in any of the organs and tissues examined. It was concluded that the no observed adverse effect level of ammonium sulfate was the 0.6% diet, which is equivalent to 256 and 284 mg/kg b.w./day in males and females, respectively, and the compound is non carcinogenic under the conditions of the study.

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Keywords: Ammonium sulfate; Chronic toxicity; Carcinogenicity; F344 rats

1. Introduction

Ammonium sulfate has traditionally been produced as a by-product when ammonia gas in coke-ovens is collected with sulfuric acid. It has been certified as a food additive in Japan, as well as the EU and US, and is chiefly used as a fermentation nutrient agent for yeast in the brewing industry (Tanimura, 1999). It has similarly found application as a major nitrogen source in winemaking and breadmaking (Cyran et al., 2003; Mendes-Ferreira et al., 2004), being more easily utilized than organic nitrogen compounds (Milstein and Bekker, 1976). Regarding general biological effects of ammo-

niun compounds in mammals, dietary administration of ammonium chloride and ammonium sulfate for 1 week or more has been reported to induce mild metabolic acidosis (Oetzel et al., 1991; Wang and Beede, 1992; Wesson, 1998). WHO has issued acute oral toxicity data for ammonium sulfate in rats, and has reported that it is not mutagenic in microbial tests with *Salmonella* and *Saccharomyces* (WHO, 1986). Inhalation studies in rats and guinea pigs with ammonium sulfate have indicated no effects on arterial blood gases or on lung volume or pathology (Pepelco et al., 1980). More recently, short-term inhalation studies of ammonium bisulfate and ammonium ferrosulfate in rats demonstrated neither cytotoxic nor histological changes in the lungs (Cassee et al., 2002). Despite the fact that ammonium sulfate is widely used as a food additive, no oral toxicity/carcinogenicity data with long-term

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administration have hitherto been available. Doses were selected for the current experiments on the basis of a 13-week dietary study in which rats were administered diets containing 0%, 0.38%, 0.75%, 1.5% and 3.0% ammonium sulfate in which reduced body weight gain and diarrhea were noted in the 3.0% males. Therefore, the no observed adverse effect level (NOAEL) of ammonium sulfate was estimated to be 1.5% (866 mg/kg/day) in males, and 3.0% (1975 mg/kg/day) in females (Takagi et al., 1999). Based on this result, 52-week chronic and 104-week carcinogenicity studies at dietary concentrations up to 3.0% were conducted to more fully evaluate the potential toxicity and/or carcinogenicity of ammonium sulfate.

2. Materials and methods

2.1. Chemicals

Ammonium sulfate, obtained from Yoneyama Kagaku Kogyo (Osaka, Japan), was mixed at concentrations of 0%, 0.1%, 0.6%, 1.5% and 3.0% into powdered basal diet CRF-1 (Oriental Yeast; Tokyo, Japan) and then pelleted. Recapture rates for ammonium sulfate from the admixed diet at each concentration level were confirmed to be 95.4–98.7%. Stability of ammonium sulfate in the solid pellet diet was evaluated and no decomposition was confirmed after storage under refrigeration or at room temperature for 2 weeks.

2.2. Experimental animals and housing conditions

A total of 190 male and 190 female F344/DuCrj rats were purchased at 5 weeks of age from Charles River Japan (Atsugi, Japan) and acclimated for 1-week prior to the commencement of the experiment. Rats were randomly allocated, 10 animals/sex/group for the chronic toxicity study and 50 animals/sex/group for assessment of carcinogenicity, and maintained in a room with a barrier system under constant conditions: temperature of $24 \pm 1^\circ\text{C}$, relative humidity $55 \pm 5\%$, ventilation frequency of 18 times/h and 12-h light/dark cycle. The animals were housed three or four rats per plastic cage, on sterilized soft wood chips (Sankyo Laboratory Service, Tokyo, Japan), and were given free access to tap water and diet.

2.3. Experimental design

The animals received ammonium sulfate at concentrations of 0% (control), 0.1%, 0.6% or 3.0% in the diet for 52 weeks for chronic toxicity and 0% (control), 1.5% or 3.0% in the diet for 104 weeks to test for carcinogenicity, the test chemical-supplemented diet being replaced regularly within 2-week intervals. During

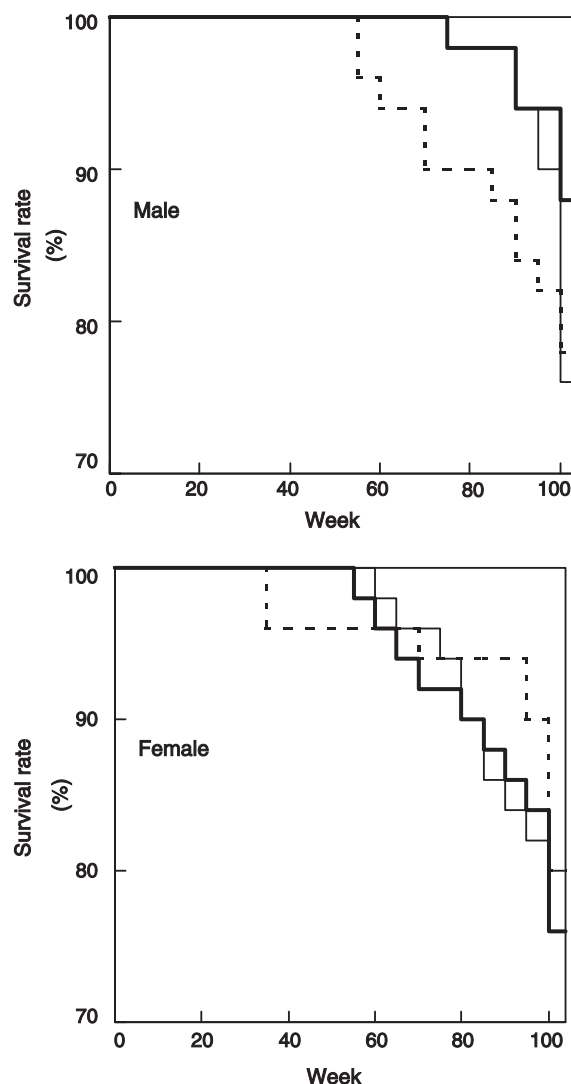


Fig. 1. Survival curves of rats dietary administered 0%, 1.5% or 3% ammonium sulfate for 104 weeks (—, 0% group; ---, 1.5% group; ···, 3.0% group).

Table 1

Final body weight, food consumption and the amount of ammonium sulfate intake in rats fed diet containing ammonium sulfate for 52 weeks

	Dose level (%)	Final body weight (g)	Food consumption (g/rat/day)	Intakes of ammonium sulfate (mg/kg b.w./day)
Male	0	410.9 $\pm$ 12.3	13.9	
	0.1	428.6 $\pm$ 17.6	13.6	42
	0.6	416.7 $\pm$ 23.7	13.4	256
	3.0	400.5 $\pm$ 15.1	15.7	1527
Female	0	207.4 $\pm$ 13.5	8.4	
	0.1	220.3 $\pm$ 8.7	8.6	48
	0.6	219.2 $\pm$ 13.6	8.4	284
	3.0	212.7 $\pm$ 24.4	8.6	1490

Each value represents the mean throughout the administration period.

Table 2

Final body weight, food consumption and the amount of ammonium sulfate intake in rats fed diet containing ammonium sulfate for 104 weeks

	Dose level (%)	Final body weight (g)	Food consumption (g/rat/day)	Intakes of ammonium sulfate (mg/kg b.w./day)
Male	0	409.6 ± 37.3	13.7	
	1.5	412.0 ± 49.6	13.6	564.1
	3.0	395.1 ± 24.1	14.8	1288.2
Female	0	266.0 ± 30.0	8.8	
	1.5	279.7 ± 22.3	9.2	649.9
	3.0	258.1 ± 27.0	9.1	1371.4

Each value represents the mean throughout the experimental period.

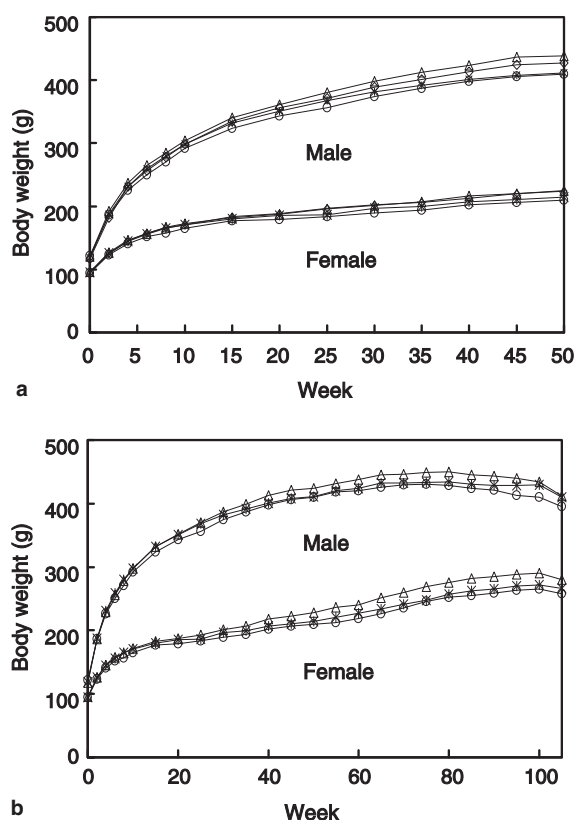


Fig. 2. (a) Body weight curves of F344 rats treated with ammonium sulfate for 52 weeks (X, 0% group; △, 0.1% group; ◇, 0.6% group; ○, 3.0% group) and (b) for 104 weeks (X, 0% group; △, 1.5% group; ○, 3.0% group).

the treatment period, clinical signs and mortality were observed daily, and body weight and food consumption were recorded every 2 weeks until week 10 and every 5 weeks thereafter. After the treatment periods of 52 and 104 weeks for the chronic and carcinogenicity studies, respectively, all survivors were euthanized under ether anesthesia by exsanguination. In the chronic toxicity study, animals were fasted overnight prior to the terminal necropsy, and blood samples were collected from

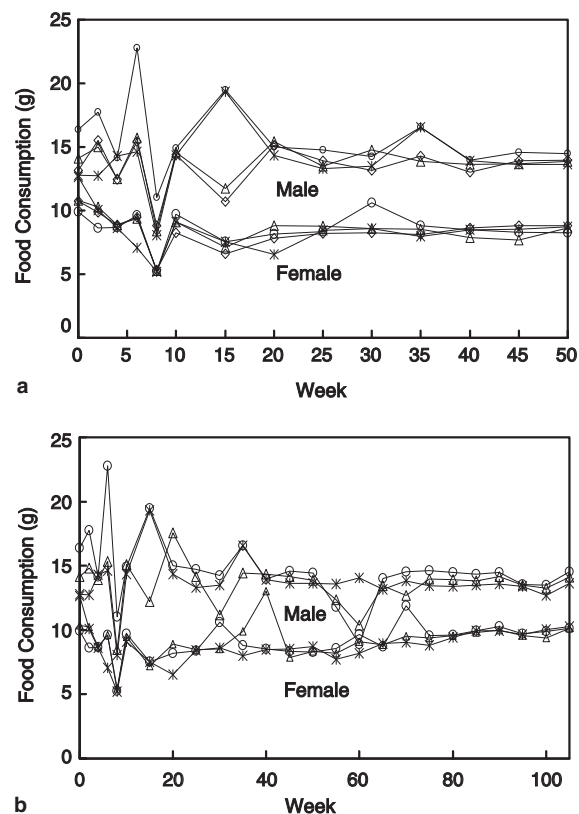


Fig. 3. (a) Food consumption changes of F344 rats treated with ammonium sulfate for 52 weeks (X, 0% group; △, 0.1% group; ◇, 0.6% group; ○, 3.0% group) and (b) for 104 weeks (X, 0% group; △, 1.5% group; ○, 3.0% group).

the abdominal aorta at the time of sacrifice. Hematological examinations were performed using an automatic hematology analyzer, M-2000 (Toa Medical Electronics; Hyogo, Japan) and the following parameters were determined: red blood cell count (RBC), hemoglobin concentration (Hb), hematocrit (Ht), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), platelet count (Plt) and white blood cell count (WBC). Differential leukocyte counts and the reticulocyte count (Ebl) were obtained with a Microx HEG-120A (Omron Tateishi Electronics; Tokyo, Japan). Serum biochemistry was performed at SRL Inc. (Tokyo, Japan) for the following 17 parameters: total protein (TP), albumin (Alb), albumin/globulin ratio (A/G), total bilirubin (T-bil), total cholesterol (T-Chol), triglyceride (TG), blood urea nitrogen (BUN), creatinine (Cre), calcium (Ca), inorganic phosphorus (IP), sodium (Na), potassium (K), chloride (Cl), aspartate transaminase (AsT), alanine transaminase (AlT), alkaline phosphatase (ALP) and γ -glutamyl transpeptidase (γ -GTP). All animals in the chronic and carcinogenicity studies were subjected to complete necropsy. Those that died or became moribund during the treatment period were also subjected to a complete necropsy as soon as they were found. Brain, lungs,

heart, spleen, liver, adrenals, kidneys and testes were weighed. As for adrenals, kidneys and testes, weights of each side were separately recorded and the total of both sides was used for calculation of group mean and SD values. In addition to these organs, the nasal cavity, trachea, aorta, pituitary, thyroids, parathyroids, salivary glands, tongue, esophagus, stomach, duodenum, jejunum, ileum, cecum, colon, rectum, pancreas, urinary bladder, epididymides, prostate, seminal vesicles, ovaries, uterus, vagina, mammary glands, skin, mesenteric and submandibular lymph nodes, thymus, sternum,

femur including bone marrow, sciatic nerve, trigeminal nerve, spinal cord (cervical, thoracic and lumbar cords), eye, Harderian gland and thigh muscle were excised and specimens from all these organs or tissues were fixed in 10% buffered formalin for paraffin-embedding sectioning and staining with hematoxylin and eosin for histopathological examination. All organs and tissues in the control and 3.0% group animals were histopathologically examined in both chronic toxicity and carcinogenicity studies. Additionally, macroscopically abnormal sites in the 0.1% and 0.6% group animals in the chronic

Table 3

Hematology in male rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

	Dose level (%)			
	0 (Control)	0.1	0.6	3.0
RBC $10^{10}/L$	841.0 $\pm$ 39.3	875.0 $\pm$ 35.5	842.0 $\pm$ 22.3	868.0 $\pm$ 40.2
Hb g/dL	13.6 $\pm$ 1.0	14.3 $\pm$ 0.7	13.9 $\pm$ 0.4	14.1 $\pm$ 0.5
Ht %	43.6 $\pm$ 1.9	45.7 $\pm$ 2.0	44.1 $\pm$ 1.0	44.9 $\pm$ 2.1
MCV fL	51.8 $\pm$ 0.6	52.2 $\pm$ 0.6	52.4 $\pm$ 0.5	51.7 $\pm$ 1.0
MCH pg	16.2 $\pm$ 0.7	16.4 $\pm$ 0.3	16.5 $\pm$ 0.3	16.2 $\pm$ 0.4
MCHC g/dL	31.3 $\pm$ 1.5	31.3 $\pm$ 0.6	31.6 $\pm$ 0.5	31.3 $\pm$ 0.6
Plt $10^{10}/L$	65.6 $\pm$ 5.7	70.3 $\pm$ 6.1	66.5 $\pm$ 4.9	65.4 $\pm$ 11.6
Ebl count/200 WBC	2.4 $\pm$ 2.0	3.9 $\pm$ 2.2	3.1 $\pm$ 1.7	2.4 $\pm$ 1.6
WBC $10^8/L$	51.8 $\pm$ 16.6	44.7 $\pm$ 13.5	46.1 $\pm$ 8.1	49.6 $\pm$ 13.5
Differential cell count (%)				
Band	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Seg	36.4 $\pm$ 4.7	44.5 $\pm$ 9.0*	43.6 $\pm$ 4.1	39.9 $\pm$ 8.2
Eosino	2.1 $\pm$ 1.3	1.9 $\pm$ 0.8	2.7 $\pm$ 2.0	1.9 $\pm$ 0.8
Baso	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.2
Lympho	61.1 $\pm$ 3.8	53.2 $\pm$ 9.0	53.4 $\pm$ 4.0*	57.8 $\pm$ 8.5
Mono	0.4 $\pm$ 0.7	0.4 $\pm$ 0.6	0.4 $\pm$ 0.5	0.5 $\pm$ 0.6

Each value represents the mean $\pm$ SD.

* Significantly different from the control at $p < 0.05$.

Table 4

Hematology in female rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

	Dose level (%)			
	0 (Control)	0.1	0.6	3.0
RBC $10^{10}/L$	770.0 $\pm$ 83.8	792.0 $\pm$ 44.3	742.0 $\pm$ 29.1	744.0 $\pm$ 40.2
Hb g/dL	13.4 $\pm$ 1.5	14.0 $\pm$ 0.8	13.0 $\pm$ 0.4	13.0 $\pm$ 1.0
Ht %	43.6 $\pm$ 4.5	44.7 $\pm$ 2.2	42.2 $\pm$ 1.5	42.3 $\pm$ 2.1
MCV fL	56.6 $\pm$ 0.3	56.5 $\pm$ 0.4	56.8 $\pm$ 0.3	56.9 $\pm$ 0.4
MCH pg	17.4 $\pm$ 0.6	17.7 $\pm$ 0.4	17.5 $\pm$ 0.6	17.5 $\pm$ 0.7
MCHC g/dL	30.7 $\pm$ 1.1	31.4 $\pm$ 0.7	30.8 $\pm$ 1.0	30.7 $\pm$ 1.3
Plt $10^{10}/L$	61.3 $\pm$ 4.3	62.6 $\pm$ 3.7	60.0 $\pm$ 2.8	60.1 $\pm$ 5.1
Ebl count/200 WBC	7.3 $\pm$ 4.2	8.3 $\pm$ 3.0	6.5 $\pm$ 3.1	8.5 $\pm$ 3.4
WBC $10^8/L$	31.1 $\pm$ 6.5	28.8 $\pm$ 3.0	24.0 $\pm$ 2.9*	28.5 $\pm$ 9.7
Differential cell count (%)				
Band	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Seg	32.4 $\pm$ 16.6	26.3 $\pm$ 6.6	27.0 $\pm$ 3.6	26.0 $\pm$ 4.0
Eosino	1.1 $\pm$ 1.0	1.5 $\pm$ 0.9	1.8 $\pm$ 1.3	1.0 $\pm$ 0.7
Baso	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Lympho	66.0 $\pm$ 16.4	71.5 $\pm$ 6.3	70.4 $\pm$ 4.7	72.4 $\pm$ 4.2
Mono	0.5 $\pm$ 0.5	0.7 $\pm$ 0.6	0.8 $\pm$ 0.5	0.6 $\pm$ 0.6

Each value represents the mean $\pm$ SD.

* Significantly different from the control at $p < 0.05$.

study and all organs and tissues of the 1.5% animals in the carcinogenicity study were also histopathologically examined.

2.4. Statistics

Variance in data for body weights, hematology, serum biochemistry and organ weights was checked for homogeneity by the Bartlett test. When the data were

homogeneous, one-way analysis of variance (ANOVA) was used. In the heterogeneous cases, the Kruskal Wallis test was applied. When statistically significant differences were indicated, Dunnett's multiple test was employed for comparison between control and treated groups. Final survival rates and the incidences of tumors and non-neoplastic lesions were compared with the Fisher's exact probability test or the Mann Whitney's *U*-test.

Table 5

Serum biochemistry in male rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

	Dose level (%)			
	0 (Control)	0.1	0.6	3.0
TP g/dL	7.3 ± 0.3	7.3 ± 0.3	7.5 ± 0.3	7.1 ± 0.3
Alb g/dL	4.8 ± 0.2	4.8 ± 0.2	4.9 ± 0.2	4.7 ± 0.3
A/G	1.9 ± 0.1	2.0 ± 0.1*	1.9 ± 0.1	2.0 ± 0.1*
T Bil mg/dL	0.1 ± 0.1	0.2 ± 0.1	0.3 ± 0.1**	0.1 ± 0.0
T Cho mg/dL	119 ± 12	127 ± 13	133 ± 15	115 ± 16
TG mg/dL	129 ± 28	155 ± 50	174 ± 57	123 ± 44
BUN mg/dL	18.9 ± 1.7	20.1 ± 1.6	18.4 ± 1.0	18.5 ± 1.5
Cre mg/dL	0.36 ± 0.03	0.33 ± 0.02*	0.32 ± 0.02**	0.36 ± 0.02
Ca mg/dL	10.6 ± 0.2	10.8 ± 0.2	10.9 ± 0.4	10.8 ± 0.5*
IP mg/dL	5.1 ± 0.4	5.0 ± 0.5	4.5 ± 0.3**	4.9 ± 0.4
Na mEq/L	142 ± 3	143 ± 2	143 ± 4	143 ± 5
K mEq/L	4.3 ± 0.3	4.5 ± 0.3	4.6 ± 0.3	4.4 ± 0.3
Cl mEq/L	102 ± 3	104 ± 2	104 ± 2	102 ± 3
AsT IU/L	88 ± 4	109 ± 20*	134 ± 45**	88 ± 21
AlT IU/L	55 ± 7	79 ± 19*	97 ± 39**	57 ± 18
ALP IU/L	354 ± 75	391 ± 43	418 ± 46	375 ± 48
γ GTP IU/L	5.0 ± 3.0	5.0 ± 1.0	7.0 ± 2.0	7.0 ± 2.0

Each value represents the mean ± SD.

* Significantly different from the control at $p < 0.05$.

** Significantly different from the control at $p < 0.01$.

Table 6

Serum biochemistry in female rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

	Dose level (%)			
	0 (Control)	0.1	0.6	3.0
TP g/dL	7.5 ± 0.3	7.7 ± 0.5	7.8 ± 0.4	7.6 ± 0.7
Alb g/dL	5.4 ± 0.3	5.5 ± 0.4	5.6 ± 0.4	5.5 ± 0.4
A/G	2.5 ± 0.2	2.6 ± 0.1	2.6 ± 0.2	2.6 ± 0.2
T Bil mg/dL	0.2 ± 0.1	0.3 ± 0.1	0.3 ± 0.2	0.4 ± 0.3
T Cho mg/dL	123 ± 6	129 ± 8	129 ± 16	134 ± 13
TG mg/dL	84 ± 24	126 ± 36	121 ± 40	126 ± 67
BUN mg/dL	21.8 ± 6.7	18.6 ± 1.2	17.1 ± 1.6*	19.1 ± 2.2
Cre mg/dL	0.36 ± 0.04	0.33 ± 0.02*	0.34 ± 0.03	0.33 ± 0.02*
Ca mg/dL	10.5 ± 0.3	10.6 ± 0.4	10.7 ± 0.2	10.7 ± 0.4
IP mg/dL	4.0 ± 0.7	4.4 ± 0.5	4.3 ± 0.5	4.1 ± 0.6
Na mEq/L	142 ± 1	144 ± 5	142 ± 1	142 ± 1
K mEq/L	4.0 ± 0.4	4.2 ± 0.2	4.1 ± 0.2	4.0 ± 0.2
Cl mEq/L	103 ± 3	104 ± 4	104 ± 1	101 ± 2
AsT IU/L	68 ± 7	63 ± 5	64 ± 5	64 ± 6
AlT IU/L	35 ± 6	36 ± 4	33 ± 2	33 ± 4
ALP IU/L	165 ± 35	150 ± 9	134 ± 19*	141 ± 21
γ GTP IU/L	< 2	< 2	< 2	< 2

Each value represents the mean ± SD.

* Significantly different from the control at $p < 0.05$.

3. Results

3.1. Survival rates

In the chronic toxicity study, no mortality was found in any groups throughout the treatment period. In the

carcinogenicity study, the survival rate of control, 1.5% and 3.0% groups were 88%, 78% and 76%, respectively, for males, and 76%, 80% and 80%, respectively, for females, and no significant difference was not observed among the groups (Fig. 1).

Table 7

Organ weights of male rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

	Dose level (%)			
	0 (Control)	0.1	0.6	3.0
Body weight (g)	410.9 ± 12.3	428.6 ± 17.6	416.7 ± 23.7	400.5 ± 15.1
Absolute (g)				
Brain	2.04 ± 0.05	2.03 ± 0.07	2.05 ± 0.05	2.04 ± 0.05
Lungs	1.20 ± 0.09	1.23 ± 0.21	1.16 ± 0.07	1.13 ± 0.06
Heart	1.09 ± 0.08	1.10 ± 0.07	1.08 ± 0.05	1.08 ± 0.07
Spleen	0.73 ± 0.05	0.72 ± 0.04	0.83 ± 0.36	0.68 ± 0.04*
Liver	9.62 ± 0.58	9.92 ± 0.73	10.26 ± 0.63	10.0 ± 0.85
Adrenals	0.03 ± 0.01	0.04 ± 0.01	0.04 ± 0.00	0.04 ± 0.00
Kidneys	2.35 ± 0.25	2.32 ± 0.11	2.42 ± 0.11	2.51 ± 0.11*
Testes	3.38 ± 0.17	3.27 ± 0.11	3.25 ± 0.25	3.29 ± 0.14
Relative (g/100 g b.w.)				
Brain	0.50 ± 0.02	0.47 ± 0.02	0.49 ± 0.04	0.51 ± 0.02
Lungs	0.29 ± 0.02	0.29 ± 0.04	0.28 ± 0.02	0.28 ± 0.01
Heart	0.26 ± 0.02	0.26 ± 0.02	0.26 ± 0.02	0.27 ± 0.01
Spleen	0.18 ± 0.01	0.17 ± 0.01	0.20 ± 0.08	0.17 ± 0.01
Liver	2.34 ± 0.13	2.31 ± 0.09	2.46 ± 0.10	2.50 ± 0.16*
Adrenals	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00
Kidneys	0.57 ± 0.07	0.54 ± 0.02	0.58 ± 0.04	0.63 ± 0.04*
Testes	0.83 ± 0.04	0.76 ± 0.03*	0.78 ± 0.07	0.82 ± 0.04

Each value represents the mean ± SD.

* Significantly different from the control at $p < 0.05$.

Table 8

Organ weights of female rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

	Dose level (%)			
	0 (Control)	0.1	0.6	3.0
Body weight (g)	207.4 ± 13.49	220.3 ± 8.68	219.2 ± 13.62	212.7 ± 24.39
Absolute (g)				
Brain	1.86 ± 0.04	1.83 ± 0.04	1.83 ± 0.05	1.82 ± 0.05
Lungs	0.82 ± 0.06	0.79 ± 0.10	0.83 ± 0.12	0.79 ± 0.05
Heart	0.65 ± 0.05	0.67 ± 0.05	0.70 ± 0.03	0.67 ± 0.05
Spleen	0.44 ± 0.04	0.44 ± 0.02	0.45 ± 0.03	0.45 ± 0.07
Liver	4.44 ± 0.26	4.66 ± 0.35	4.69 ± 0.40	4.89 ± 0.42
Adrenals	0.04 ± 0.00	0.04 ± 0.01	0.04 ± 0.01	0.04 ± 0.01
Kidneys	1.25 ± 0.07	1.35 ± 0.08*	1.35 ± 0.09	1.39 ± 0.08**
Relative (g/100 g b.w.)				
Brain	0.90 ± 0.05	0.83 ± 0.04	0.84 ± 0.05	0.86 ± 0.09
Lungs	0.39 ± 0.04	0.36 ± 0.05	0.38 ± 0.06	0.37 ± 0.04
Heart	0.31 ± 0.02	0.31 ± 0.03	0.32 ± 0.03	0.32 ± 0.02
Spleen	0.21 ± 0.03	0.20 ± 0.01	0.21 ± 0.02	0.21 ± 0.03
Liver	2.15 ± 0.17	2.11 ± 0.13	2.14 ± 0.18	2.31 ± 0.18
Adrenals	0.02 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.02 ± 0.00
Kidneys	0.60 ± 0.01	0.61 ± 0.04	0.61 ± 0.06	0.66 ± 0.05

Each value represents the mean ± SD.

* Significantly different from the control at $p < 0.05$.

** Significantly different from the control at $p < 0.01$.

3.2. General condition, body weight, food consumption and chemical intake

Regarding general conditions, no obvious findings, including diarrhea (which was noted at 3.0% in the pre-

liminary 13-week study), were observed in any group in either the chronic toxicity or carcinogenicity studies. No test substance-related change in the body weights or food intake was found, except for a tendency for increase of food intake in the male 3.0% group in the chronic toxicity study (Tables 1 and 2, Figs. 2 and 3). The actual daily intakes of the test substance showed a

Table 9

Histopathological findings in male rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

Organs	Lesions	Grade	Dose level (%)	
			0 (Control)	3.0
Liver	Bile duct proliferation	+	9	8
		++	1	2
	Necrosis, focal	+	1	4
Pancreas	Acinar cell atrophy, focal	+	4	2
Kidney	Basophilic tubules	+	1	2
	Chronic nephropathy	+	0	1
Heart	Myocarditis, focal	+	6	6
Tongue	Mononuclear cell infiltration	+	1	0
Rectum	Erosion	+	1	0
Pituitary	Basophilic cell hyperplasia, diffuse	+	1	0
Adrenal	Pheochromocytoma, malignant		0	1
Prostate	Luminal dilatation	+	5	6
Spinal cord	Calcification	+	1	1
Adipose tissue in abdominal cavity	Necrosis, focal	+	1	1

Table 10

Histopathological findings in female rats fed diet containing ammonium sulfate for 52 weeks (number of animals = 10)

Organs	Lesions	Grade	Dose level (%)	
			0 (Control)	3.0
Liver	Bile duct proliferation	+	0	3
	Granuloma	+	3	2
	Altered hepatocellular foci	+	7	9
Spleen	Increased extramedullary hematopoiesis	+	1	0
Pancreas	Acinar cell atrophy, focal	+	1	0
Heart	Myocarditis, focal	+	1	2
Tongue	Mononuclear cell infiltration	+	1	2
Pituitary	Cyst	+	1	2
	Adenoma, anterior lobe		0	2
Thyroid	C cell hyperplasia		1	0
Uterus	Endometrial stromal polyp		1	0
Harderian gland	Mononuclear cell infiltration	+	3	0

Table 11

Histopathological findings in male rats fed diet containing ammonium sulfate for 104 weeks (non neoplastic lesions)

Organs	Lesions	Dose level (%)		
		0 (Control) [50]	1.5 [48]	3.0 [50]
Liver	Altered hepatocellular foci	37	34	38
	Bile duct proliferation	39	38	42
	Necrosis, focal	2	5	4
	Cholangiofibrosis	1	0	0
	Calcification	1	0	0
Spleen	Increased extramedullary hematopoiesis	9	7	10
Kidney	Chronic nephropathy	19	29*	28
	Calcification	1	0	0
Pancreas	Islet cell hyperplasia	1	1	4
	Basophilic cell foci	0	1	0
Lung	Alveolar hyperplasia	0	1	1
Eye	Retinal atrophy	25	30	36
Thyroid	C cell hyperplasia	9	4	11
	Cyst	2	0	0
Mammary gland	Cyst	2	0	4
	Cyst, multiple	1	1	2
Adrenal	Fatty change, cortex	0	1	2
	Hyperplasia, cortex	6	1	3
	Hyperplasia, medulla	10	11	15
Pituitary	Cyst, anterior lobe	4	5	3
	Hyperplasia, anterior lobe	5	8	8
	Cyst, posterior lobe	1	0	1
	Dilated Rathke's cleft	3	1	0
	Hyperplasia, intermediate lobe	1	0	0
Jejunum	Diverticulum	0	0	1
Testis	Interstitial cell hyperplasia	2	2	1
Prostate	Inflammation	1	1	4
Urinary bladder	Transitional cell hyperplasia	3	2	0
Nasal cavity	Chronic inflammation, nasolacrimal duct	2	0	2
Spinal cord	Gliosis	1	0	0
	Calcification	0	1	0
Bone marrow	Increased hematopoiesis	7	3	8
	Myelofibrosis	1	0	1

Number of animals are given in "[]".

* Significantly different from the control at $p < 0.05$.

Table 12

Histopathological findings in female rats fed diet containing ammonium sulfate for 104 weeks (non neoplastic lesions)

Organs	Lesions	Dose level (%)		
		0 (Control) [48]	1.5 [48]	3.0 [50]
Liver	Altered hepatocellular foci	41	42	44
	Bile duct proliferation	35	41	39
	Cyst, bile duct	0	0	1
	Microgranuloma	11	12	9
	Necrosis, focal	1	3	3
	Fatty degeneration	1	1	1
Spleen	Increased extramedullary hematopoiesis	1	4	4
Kidney	Chronic nephropathy	1	5	3
	Vacuolar degeneration, proximal tubule	0	1	0
	Cyst	0	2	0
	Dilation, pelvis	0	2	1
Pancreas	Islet cell hyperplasia	0	1	0
	Basophilic cell foci	0	1	0
Tongue	Squamous cell hyperplasia	1	0	0
Heart	Calcification	1	0	0
Salivary gland	Hyperplasia, focal	1	3	1
	Granuloma	1	0	0
Brain	Dilation, ventricle	2	0	0
Lung	Alveolar hyperplasia	3	3	0
	Granuloma	0	1	1
Eye	Retinal atrophy	31	34	36
	Retinitis	0	1	0
Harderian gland	Granuloma	1	0	0
	Chronic inflammation	0	1	0
Thyroid	C cell hyperplasia	22	27	21
	Cyst, ultimobranchial	0	0	1
Parathyroid	Hyperplasia, nodular	0	1	0
Mammary gland	Cyst	1	0	0
Adrenal	Fatty change, cortex	17	20	15
	Hyperplasia, cortex	4	4	9
	Hyperplasia, medulla	1	1	0
	Angiectasis	3	1	2
Pituitary	Cyst, anterior lobe	19	21	23
	Hyperplasia, anterior lobe	8	12	14
	Cyst, Rathke's cleft	7	7	8
	Angiectasis	2	1	1
Stomach	Intestinal metaplasia	3	0	0
	Edema, forestomach	0	1	0
	Hyperplasia, forestomach	0	1	2
Ovary	Atrophy	5	3	6
Uterus	Cystic hyperplasia	18	18	17
	Endometrial hyperplasia	2	1	3
	Adenomyosis	1	0	1
	Endometritis	0	2	1
Vagina	Cyst, submucosa	0	0	1

Table 12 (continued)

Organs	Lesions	Dose level (%)		
		0 (Control) [48]	1.5 [48]	3.0 [50]
Nasal cavity	Chronic inflammation, nasolacrimal duct	19	17	14
	Necrosis, olfactory epithelium	1	0	0
Bone marrow	Increased hematopoiesis	1	5	3
	Myelofibrosis	0	1	1

Number of animals are given in “[]”.

good correlation with the expected dose (Tables 1 and 2).

3.3. Hematology and serum biochemistry

No significant variation was found in erythrocytic parameters and Plt among the groups in the chronic toxicity study. Some slight changes were found in WBC parameters but there was no dose-relation and these were therefore considered to be incidental (Tables 3 and 4). No dose-related alteration was found in any of the serum biochemical parameters in the chronic toxicity study (Tables 5 and 6).

3.4. Organ weights

Absolute and relative kidney weights were increased or showed a tendency for increase at 3.0% in both sexes in the chronic toxicity study. Absolute spleen weights were decreased and relative liver weights were increased in the 3.0% male dose group. No dose-related changes were found in the other organs (Tables 7 and 8).

3.5. Necropsy and histopathology

There were no obvious macroscopic findings in any group in either the chronic toxicity or carcinogenicity studies, except for massive, nodular or focal lesions suggesting neoplastic change in the carcinogenicity study. Histopathologically, in the chronic toxicity study, several non-neoplastic lesions, such as bile duct proliferation in the liver and focal myocarditis in the heart were noted in the control and 3.0% group, and there were no significant differences in their incidences between the groups in either sex. As neoplastic lesions, malignant pheochromocytoma of the adrenal in the male 3.0% group, two adenomas in the anterior pituitary in females of the 3.0% group, and uterine endometrial stromal polyp in a female control rat were noted (Tables 9 and 10). In the carcinogenicity study, as a non-neoplastic lesion, incidence of chronic nephropathy in the kidney was significantly increased in the male 1.5% group. As for other lesions, such as altered hepatocellular foci

Table 13
Histopathological findings in male rats fed diet containing ammonium sulfate for 104 weeks (neoplastic lesions)

Organs	Lesions	Dose level (%)		
		0 (Control) [50]	1.5 [48]	3.0 [50]
Liver	Hepatocellular adenoma	3	0	1
	Hepatocellular carcinoma	0	2	0
Spleen	Hemangiosarcoma	1	0	0
Kidney	Lipoma	1	0	0
	Nephroblastoma	0	1	0
Pancreas	Adenoma, islet cell	2	1	2
Heart	Fibroma	0	0	1
Lung	Adenoma	2	4	1
	Adenocarcinoma	0	1	2
Eye	Melanoma, amelanotic	1	0	0
Thyroid	Adenoma, C cell	11	12	10
	Adenocarcinoma, C cell	4	1	3
Skin	Basal cell tumor	0	0	1
	Trichoepithelioma	1	0	0
Mammary gland	Fibroadenoma	0	2	4
	Adenoma	1	0	0
	Adenocarcinoma	0	1	0
Adrenal	Pheochromocytoma	7	6	4
	Pheochromocytoma, malignant	1	5	1
	Adenoma, cortical	1	0	0
Pituitary	Adenoma, anterior lobe	11	8	6
	Adenocarcinoma, anterior lobe	2	3	4
Jejunum	Adenocarcinoma	0	0	1
Testis	Interstitial cell tumor	45	41	46
Prostate	Adenoma	11	5	5
	Leiomyosarcoma	1	0	0
Urinary bladder	Papilloma	1	0	0
	Leiomyoma	0	0	1
Bone marrow	Leukemia, myeloid	0	1	0
Systemic	Leukemia, large granular lymphocytic	6	4	6
Others	Osteosarcoma	1	0	0
	Adenoma, preputial glands	3	3	3
	Carcinoma, preputial glands	0	1	0
	Carcinoma, Zymbal's gland	1	0	0

Number of animals are given in “[]”.

and bile duct proliferation in the liver and retinal atrophy in the eye were noted in the control and treatment groups, with no significant inter-group difference in their incidences or severity (Tables 11 and 12). Neoplastic lesions such as C-cell adenomas/adenocarcinomas in the thyroids, fibroadenomas/adenomas/adenocarcinomas

in the mammary glands, adenomas/adenocarcinomas in the pituitary glands, interstitial cell tumors in the testes and endometrial stromal polyps in the uteri were noted in all groups, but these are all known to occur spontaneously in this strain, and neither increases in their incidences nor specific types of lesions were observed in the test-substance-administered groups (Tables 13 and 14).

Table 14
Histopathological findings in female rats fed diet containing ammonium sulfate for 104 weeks (neoplastic lesions)

Organs	Lesions	Dose level (%)		
		0 (Control) [48]	1.5 [48]	3.0 [50]
Spleen	Hemangiosarcoma	1	0	0
	Sarcoma, NOS	0	1	0
Kidney	Lipoma	1	1	0
Pancreas	Adenoma, islet cell	0	3	0
Lung	Adenoma	0	0	1
	Adenocarcinoma	1	0	0
	Sarcoma, NOS	0	0	1
Thyroid	Adenoma, C cell	8	11	8
	Adenocarcinoma, C cell	1	1	1
	Adenocarcinoma, follicular cell	0	1	0
Skin	Basal cell tumor	1	1	0
	Trichoepithelioma	1	0	0
	Carcinoma, sebaceous	1	0	0
Mammary gland	Fibroadenoma	7	14	7
	Adenocarcinoma	0	1	0
Adrenal	Pheochromocytoma	0	0	1
Pituitary	Adenoma, anterior lobe	14	6	11
	Adenocarcinoma, anterior lobe	5	10	7
	Adenocarcinoma, intermediate lobe	1	0	0
Ovary	Tumor, stromal cell	1	0	1
	Tumor, Sertoli cell	0	1	0
Uterus	Adenoma, endometrial	0	1	0
	Adenocarcinoma, endometrial	1	1	1
	Polyp, endometrial stromal	9	16	11
	Sarcoma, endometrial stromal	0	1	2
	Leiomyoma	0	1	0
Vagina	Sarcoma	0	1	1
Clitoral gland	Adenoma	2	1	1
	Carcinoma	0	0	1
Systemic	Leukemia, large granular lymphocytic	7	6	2
	Lymphoma	2	0	1
	Malignant fibrous histiocytoma	0	0	1

Number of animals are given in “[]”.

4. Discussion

The present 52-week chronic and 104-week carcinogenicity dietary studies of ammonium sulfate in Fischer 344 rats did not demonstrate major toxicological effects including reduced body weight gain and diarrhea, which were observed in the male 3.0% group in the previous 13-week subchronic toxicity study (Takagi et al., 1999). There is a possibility that the present 3.0% did not reach to the maximum tolerated dose, while in a preliminary 2-week rat palatability study 5.0% group males showed significant reduction of body weight gain and diarrhea from the first week, but not their 2.5% counterparts (unpublished data). Therefore, the dose of 3.0% was considered to be critical for these findings. A tendency for increase in food intake but no body weight change was observed in the male 3.0% group in the chronic toxicity study. Further study is needed to clarify whether metabolic effects, e.g., influences on digestive fluid secretion and/or digestive enzyme (Olivera et al., 2003), might be affected or not by the ammonium sulfate treatment to rats.

There was no dose-related change in hematology and serum biochemistry in the chronic toxicity study. Absolute and relative kidney weights were increased in both sexes of the 3.0% groups in the chronic toxicity study, and similar results were noted in the previous 13-week subchronic toxicity study (Takagi et al., 1999). In the chronic toxicity study, no significant histopathological renal change was noted. On the other hand, by 104 week, the incidence of chronic nephropathy was increased in the 1.5% and 3.0% males as compared to the control, and relation to the kidney weight changes seemed not to be negligible. Moreover, relative liver weights were significantly increased in 3.0% males, and showed a tendency for increase in 3.0% females in the present chronic study. The results suggest that high dose ammonium sulfate influences some hepatic functions, but this may not be toxicologically significant, given the lack of relevant biochemical and histopathological changes. The main pathway for the detoxification of ammonia is well known as the urea cycle, and there are other routes for ammonia utilization in the liver of ureotelic animals, whereby ammonia is used to form metabolites such as glutamate and glutamine (Masumi et al., 1999). These reactions might be involved.

In the carcinogenicity study, while several tumors were noted there were no significant inter-group differences in their incidence or histological types, all being spontaneous tumors described previously in the same strain (Maekawa et al., 1983). Although some neoplastic lesions, such as malignant pheochromocytoma in the adrenal gland and adenomas in the anterior pituitaries were observed in the 3.0% group in the chronic toxicity study, no induction of such tumors was evident in the carcinogenicity study.

Ammonium sulfate is used at concentration of 0.1–0.4% in the flour and 0.5–1% in the yeast mash for bread making and brewing, respectively (Tanimura, 1999). The exposure levels in the present experiments were estimated much greater than those of human. Thus, risk of weak liver or kidney weight changes for humans may be negligible.

In conclusion, the no observed adverse effect level (NOAEL) for ammonium sulfate was estimated from the chronic toxicity study of F344 rats to be 0.6% in both sexes, which is equivalent to 256 and 284 mg/kg/day in males and females, respectively, and no evidence of long-term carcinogenic activity was found in the carcinogenicity study.

Acknowledgements

This study was supported by a Grant-in-Aid for Research on Food Sanitation from the Ministry of Health, Labour and Welfare of Japan.

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